Srikant Kharat

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PROFESSIONAL SUMMARY

C++ software engineer with **4 years 8 months** at Tata Consultancy Services developing automotive **diagnostic middleware and ECU reprogramming software**, currently working onsite in Kanagawa, Japan. Hands-on with diagnostic communication over **UDS (ISO 14229)**, **CAN / ISO-TP (ISO 15765)** and **DoIP**, driven through **SAE J2534 PassThru** and **D-PDU API (ISO 22900-2)** vehicle interfaces. Implements and debugs diagnostic request/response handling, ECU reprogramming workflows and communication error recovery; resolves production field defects through protocol trace and communication log analysis. Seeking a hands-on C++ development role in Japan or Europe.

TECHNICAL SKILLS

Programming Languages: **C++** (primary), **C#** **[production or personal only?]**, Python, SQL

C++ Standards & Language Features: **[confirm which: C++98/03, C++11, C++14, C++17, C++20]** - object-oriented design, RAII, templates, smart pointers, STL containers & algorithms, exception handling, **[move semantics / lambdas / concurrency - keep only what you have used]**

C++ Standard Library (production use): **[strike anything you have NOT used at work:]** std::vector, std::map, std::unordered_map, std::string, std::unique_ptr, std::shared_ptr, std::thread, std::mutex, std::lock_guard, std::condition_variable, std::chrono, std::function, std::optional

Automotive Diagnostics & Standards: **UDS (ISO 14229)**, **CAN / ISO-TP (ISO 15765)**, **DoIP**, **D-PDU API (ISO 22900-2)**, **SAE J2534 PassThru**, ECU reprogramming & programming, DTC, DID, diagnostic routines, ISO/OSI layer-level protocol debugging; **[ODX (ISO 22901) / OTX (ISO 13209) / CAN-FD / D-Server (ISO 22900-3) - confirm which you have actually worked with]**

Tools & Platforms: Visual Studio 2005 / 2019 / 2022 (MSVC), **Vector CANoe**, **Wireshark**, Bosch VI3 VCI, Git, Docker, Windows, Linux; **[CMake / GoogleTest / Valgrind / AddressSanitizer / SonarQube / vTESTstudio - confirm which your project uses]**

Software Engineering: Object-oriented design, design patterns **[which ones?]**, multithreading & synchronization, memory & lifetime management, debugging and root-cause analysis, bench and in-vehicle validation, communication log & trace analysis, code reviews, data structures & algorithms, system design; **[unit / integration testing - confirm whether you write automated tests]**

PROFESSIONAL EXPERIENCE

Onsite Technical Engineer - Tata Consultancy Services | Mar 2025 - Present | Kanagawa, Japan

Multi-Vehicle Reprogramming Tool - C++ Developer (onsite)

- Implement reprogramming support for new vehicle models in the production **C++** application: per-ECU flash sequence, data block transfer handling, erase and checksum routines, and session timing / keep-alive handling.
- Debug field reprogramming failures at protocol level - message segmentation and flow control, response-pending handling, session timeouts, DoIP routing - from **Vector CANoe** traces and **Wireshark** captures, and fix the defects in the diagnostic communication layer.
- Validate actual ECU behaviour against the specification on vehicle and bench setups together with the customer's engineers in Japan, and turn confirmed behaviour into implementation-ready designs for the development team.
- **[Do you review the offshore team's code / write their technical design? keep this bullet only if yes]**

System Engineer - Tata Consultancy Services | Apr 2023 - Feb 2025 | India

Multi-Vehicle Reprogramming Tool - C++ Developer & Tool Owner

- Developed and maintained the **C++** reprogramming application end to end, implementing per-model, per-ECU reprogramming requirements and the sequence logic that reprograms **multiple ECUs of a vehicle in a single session**.
- Implemented **UDS**-based diagnostic request handling, response validation and communication error recovery - retry, timeout and negative-response handling - for the reprogramming workflows.
- **[Did you implement the VCI integration layer yourself?]** Integrated the application with **SAE J2534 PassThru** and **D-PDU API (ISO 22900-2)** vehicle interfaces so the same reprogramming sequence runs across different interface hardware.
- Built **bench ECU test setups and wiring harnesses** to reproduce reported reprogramming failures off-vehicle; validated every fix on bench and in vehicle before release.
- Analysed communication logs and diagnostic traces to isolate protocol-level and application defects reported from the field, and implemented the corresponding fixes.

- Tool reprogrammed **5,000+ vehicles in a single year at under 2% error rate**; received the **On-the-Spot (Team) Award**.

Assistant System Engineer - Tata Consultancy Services | Feb 2022 - Mar 2023 | India

Consult-III-Plus (after-sale dealer diagnostic application) - Middleware (MW) Developer

- Core developer in the **middleware team** - the layer between the diagnostic application and the vehicle communication interface - of a diagnostic application used in dealer workshops.
- Implemented **UDS** diagnostic request/response handling, **ECU reprogramming and programming flows**, and diagnostic services including DID read/write and DTC read/clear, in **C++**.
- Investigated market issues reported from the field: reproduced the fault on bench, performed **root-cause analysis** across the middleware stack, implemented the fix and validated it on bench and vehicle.
- Supported diagnostic requirement analysis with the diagnostics team and validated the implemented services against the specification.

EDUCATION & CERTIFICATIONS

B.E., Computer Science & Engineering - Mahatma Gandhi Mission College of Engineering, Nanded, India | **[start year]** - 2021 **[| GPA ?]**

Japanese-Language Proficiency Test (JLPT) **[level]** | **[year - passed or targeting?]**

Automotive Diagnostics & ECU Reprogramming - Tata Consultancy Services | 2023

C++ Programming (Professional) - Tata Consultancy Services | 2024

ACHIEVEMENTS

- **On-the-Spot (Team) Award**, Tata Consultancy Services - for delivery of the Multi-Vehicle Reprogramming Tool (5,000+ vehicles reprogrammed in one year, under 2% error rate).
- Selected for **onsite deployment to Japan** as the technical interface between the customer's engineering team and offshore development.